

Medical AI for Doctors

Prompting Guide for Physician-Builders

How doctors can get more precise, useful outputs from AI systems

Series 3 of 3

Clear promise: A guide to writing better prompts for clinical, educational, and technical work.

Part of the Doctors Who Code lead magnet series for physician-builders.

Guide 3 of 3

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How doctors can get more precise, useful outputs from AI systems

Clear promise: A guide to writing better prompts for clinical, educational, and technical work.

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What a prompt really does

Prompting is not magic. It is structured communication. A prompt tells the system what problem it is solving, what boundaries matter, and what shape the output should take.

Physicians already do this in other forms. We frame consult questions, write orders, define tasks for trainees, and communicate with constraints. Prompting extends that habit into human-AI interaction.

The anatomy of a strong prompt

A strong prompt usually includes four elements: context, task, constraints, and output format. Context tells the system what it is working with. Task tells it exactly what to do. Constraints define boundaries such as tone, audience, length, and exclusions. Output format tells it how the answer should be organized.

Specificity improves usefulness. The more clearly you define the work, the less the model has to guess.

A weak prompt and a stronger prompt

- Weak: Write something about gestational diabetes.
- Stronger: Write a patient handout on gestational diabetes for an eighth-grade reading level. Use short paragraphs, explain the condition, why it matters, and what lifestyle steps may help. Avoid jargon. End with three questions patients can ask their doctor.

Prompt patterns for physicians

- Summarize this chart for a follow-up visit and separate historical facts from active concerns.
- Draft a patient portal reply in plain language and keep the tone calm, clear, and bounded.
- Turn these bullet points into a structured handout for residents with sections for diagnosis, risk, management, and teaching pearl.

Prompt patterns for coding and app ideas

- Explain this codebase to a physician beginner and define the front end, back end, and data flow separately.
- Turn this workflow pain point into a small app concept with user, input, output, and first-version scope.
- Generate a checklist for validating a guideline-based calculator before publishing it.

Common mistakes and how to fix them

- Being too vague about the task.
- Forgetting to define the audience.

- Leaving out exclusions and safety boundaries.
- Requesting an answer format that is hard to review.
- Treating the first output as final instead of iterative.

Closing thought

Physicians already think in structured frameworks. Prompting simply extends that discipline into AI-assisted work. The better you define the problem, the better the system can help.